Excess Entropy and Self-Diffusion in Multicomponent Charged Fluids: A Comparative Integral Equation and Molecular Dynamics Investigation

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We investigate the excess entropy, structural two-body entropy, and transport properties of multicomponent charged Lennard-Jones mixtures along a supercritical isotherm ($T^* = 2.00$, $\Gamma = 1.0$) up to dense liquid packing ($\rho^* = 0.70$). We contrast a symmetric 3-component electrolyte mixture (neutral solvent with a 2:2 salt, ionic strength $I = 0.40$) against an asymmetric 5-component mixture (neutral solvent with mixed 1:1 and 2:2 salts at identical total packing fraction $\eta \approx 0.3665$, but lower ionic strength $I = 0.19$). Hypernetted Chain (HNC) integral equation theory incorporating Allnatt electrostatic screening is evaluated alongside GPU-accelerated molecular dynamics simulations. We demonstrate that numerical differentiation of the HNC excess free energy functional rigorously reproduces the virial compressibility factor ($Z_{\rm diff} \equiv Z_{\rm virial}$) across the entire density range, confirming internal Euler thermodynamic consistency. Structural entropy analysis demonstrates that the two-body Green-Wallace pair entropy $s^{(2)}$ accounts for over 96% of the total excess entropy $s_{\rm ex}$ and captures 91.1% of the entropic difference between the two mixtures ($\Delta s_{\rm ex} = -0.0192\, k_B$ versus $\Delta s^{(2)} = -0.0175\, k_B$). Self-diffusion coefficients evaluated from center-of-mass corrected mean-squared displacements reveal that the enhanced Coulombic caging and tighter ionic pairing in the 3-component system retard divalent ion mobility ($D_{z2}^{3C}/D_{z2}^{5C} = 0.954$). Crucially, the relation between diffusion and excess entropy quantitatively satisfies Rosenfeld's excess entropy scaling relation, yielding a scaling exponent $B \approx +2.44$ for divalent ions ($B^{(2)} \approx +2.67$ using $s^{(2)}$). These findings establish the predictive capacity of excess entropy scaling in complex multicomponent electrolytes.

I. INTRODUCTION

Understanding the equilibrium thermodynamics, structural organization, and transport dynamics of multicomponent charged fluid mixtures represents a cornerstone of liquid state physics and physical chemistry [1, 2]. Electrolyte solutions and ionic liquids exhibit intricate structural correlations arising from the competition between short-range excluded-volume repulsions, dispersive attractions, and long-range Coulomb interactions [3, 4]. While classical Debye-Hückel theory and its mean-spherical extensions provide reasonable approximations in the dilute regime, dense electrolyte fluids require non-perturbative integral equation theories or explicit molecular simulations to capture multi-ion clustering and screening phenomena [5, 6].

A prominent framework bridging equilibrium liquid structure and nonequilibrium transport is the excess entropy scaling hypothesis formulated by Rosenfeld [7, 8]. Rosenfeld postulated that appropriately reduced transport coefficients—such as the dimensionless self-diffusion coefficient D^*, shear viscosity η^*, and thermal conductivity λ^*—depend quasi-universally on the thermodynamic excess entropy $s_{\rm ex} = s - s_{\rm id}$, via

$$D^* = A \exp\left(B \frac{s_{\rm ex}}{k_B}\right), \tag{1}$$

where A and B are system-dependent or quasi-universal scaling parameters of order unity [9, 10]. Because $s_{\rm ex} < 0$ quantifies the loss of accessible phase space relative to an ideal gas due to interparticle interactions, Eq. (1) provides a direct physical link between structural ordering and microscopic mobility.

Furthermore, within the multiparticle expansion developed by Green [11] and Wallace [12], the excess entropy can be decomposed into successive two-body, three-body, and higher-order correlation contributions,

$$s_{\rm ex} = s^{(2)} + s^{(3)} + \dots \tag{2}$$

In simple and moderately coupled liquids, the two-body pair entropy $s^{(2)}$, computed exclusively from the pair distribution functions $g_{ij}(r)$, frequently accounts for 85–95% of the total excess entropy [13–15]. However, the degree to which this two-body dominance holds in multicomponent mixtures featuring asymmetric charges and disparate valence states remains an open question.

In this work, we present a systematic study comparing two multicomponent charged Lennard-Jones fluids along a supercritical isotherm ($T^* = 2.00$, coupling parameter $\Gamma = 1.0$) up to dense liquid packing ($\rho^* = 0.70$):

1. A **3-component mixture (3C)** consisting of a neutral solvent ($x_1 = 0.80$) and a symmetric 2:2 electrolyte ($x_2 = x_3 = 0.10$, $q_2 = +2, q_3 = -2$), possessing an ionic strength $I = 0.40$.

2. A **5-component mixture (5C)** containing the same solvent fraction ($x_1 = 0.80$) and identical total packing fraction, but with a mixed 1:1 and 2:2 salt distribution ($q_2 = +1, q_3 = +2, q_4 = -1, q_5 = -2$), yielding a substantially reduced ionic strength $I = 0.19$.

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By combining the Hypernetted Chain (HNC) integral equation closure with Allnatt electrostatic screening and extensive GPU-accelerated molecular dynamics (MD) simulations, we address three central objectives: (i) assess the internal thermodynamic consistency between the HNC free energy functional and the virial equation of state; (ii) isolate how charge asymmetry and ionic valence govern the partitioning between total excess entropy $s_{\rm ex}$ and structural pair entropy $s^{(2)}$; and (iii) test whether Rosenfeld's excess entropy scaling quantitatively governs the self-diffusion of multivalent ions in multicomponent electrolyte mixtures.

II. MODEL AND THEORETICAL FRAMEWORK

A. Interaction Potentials and Mixture Specifications

We consider a classical fluid mixture of p components characterized by number densities $\rho_i = N_i/V$ $(i = 1, \ldots, p)$, with total number density $\rho = \sum_{i=1}^{p} \rho_i$ and mole fractions $x_i = \rho_i/\rho$. The particles interact via a cut-and-shifted Lennard-Jones (LJ) core augmented by Coulomb interactions:

$$u_{ij}(r) = u_{ij}^{\rm LJ}(r) + \frac{q_i q_j e^2}{4\pi\varepsilon_0\varepsilon_r r}, \tag{3}$$

where the short-range LJ contribution is truncated and shifted at $r_c = 2.5\,\sigma$:

$$u_{ij}^{\rm LJ}(r) = \begin{cases} 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right] - u_{ij}^{\rm shift}, & r \le r_c, \\ 0, & r > r_c. \end{cases} \tag{4}$$

All species share equivalent Lennard-Jones parameters: $\sigma_{ij} = \sigma$, $\varepsilon_{ij} = \varepsilon$, and mass $m_i = m$. Reduced Lennard-Jones units are employed throughout: length $r^* = r/\sigma$, density $\rho^* = \rho\sigma^3$, temperature $T^* = k_B T/\varepsilon$, energy $u^* = u/\varepsilon$, and time $t^* = t/\tau$ with $\tau = \sigma\sqrt{m/\varepsilon}$.

The electrostatic interaction strength is parametrized by the dimensionless Bjerrum length $\lambda_B^* = e^2/(4\pi\varepsilon_0\varepsilon_r\sigma k_B T)$. We define the Coulomb coupling parameter $\Gamma = \lambda_B^*/\sigma = 1.0$ at temperature $T^* = 2.00$ ($\beta^* = 1/T^* = 0.50$). The ionic strength of the mixture is given by

$$I = \frac{1}{2}\sum_{i=1}^{p} x_i q_i^2. \tag{5}$$

The Debye screening parameter κ is defined as

$$\kappa = \sqrt{4\pi\beta^*\rho^*\sum_{i=1}^{p} x_i q_i^2} = \sqrt{8\pi\beta^*\rho^* I}. \tag{6}$$

The stoichiometric compositions of the two systems investigated are summarized in Table I. Both systems share an identical solvent mole fraction ($x_1 = 0.80$, $q_1 = 0$) and identical total density $\rho^* = 0.70$, ensuring identical volume fractions $\eta = \frac{\pi}{6}\rho^* \approx 0.3665$. However, the 3C mixture exhibits a significantly higher ionic strength ($I = 0.40$) compared to the 5C mixture ($I = 0.19$), leading to substantially distinct electrostatic screening lengths: $\kappa^{-1} \approx 0.754\,\sigma$ for 3C versus $\kappa^{-1} \approx 1.094\,\sigma$ for 5C.

TABLE I. Composition, partial charges q_i, mole fractions x_i, particle numbers N_i ($N = 2000$), ionic strength I, and Debye screening length κ^{-1} at $\rho^* = 0.70, T^* = 2.00$ ($\Gamma = 1.0$).

System	Component (i)	q_i	x_i	N_i	I	$\kappa^{-1}[\sigma]$
3C	1 (Solvent)	0	0.80	1600	0.40	0.754
	2 (Cation)	+2	0.10	200		
	3 (Anion)	−2	0.10	200		
5C	1 (Solvent)	0	0.80	1600	0.19	1.094
	2 (Cation 1)	+1	0.05	100		
	3 (Cation 2)	+2	0.04	80		
	4 (Anion 1)	−1	0.09	180		
	5 (Anion 2)	−2	0.02	40		

B. Ornstein-Zernike Equations and HNC Closure

The microscopic structure of the multicomponent fluid is described by the matrix of total correlation functions $h_{ij}(r) = g_{ij}(r) - 1$ and direct correlation functions $c_{ij}(r)$, coupled via the multicomponent Ornstein-Zernike (OZ) relations [1]:

$$h_{ij}(r) = c_{ij}(r) + \sum_{k=1}^{p}\rho_k \int c_{ik}(|\mathbf{r} - \mathbf{r}'|)h_{kj}(r')\,d\mathbf{r}'. \tag{7}$$

In Fourier space, defining $\tilde{h}_{ij}(k)$ and $\tilde{c}_{ij}(k)$ via standard three-dimensional Fourier transforms, Eq. (7) takes the matrix form

$$\tilde{\mathbf{H}}(k) = \tilde{\mathbf{C}}(k) + \tilde{\mathbf{C}}(k)\mathbf{P}\tilde{\mathbf{H}}(k), \tag{8}$$

where $\mathbf{P}_{ij} = \rho_i\delta_{ij}$ is the diagonal density matrix.

To treat the long-range Coulomb tail without numerical truncation artifacts, we employ the Allnatt screening decomposition [3, 5]. The pair potential is split into short-range (SR) and long-range (LR) parts:

$$u_{ij}(r) = u_{ij}^{\rm SR}(r) + u_{ij}^{\rm LR}(r), \tag{9}$$

with $u_{ij}^{\rm LR}(r) = q_i q_j \Gamma\psi(r)$, where $\psi(r) = \mathrm{erf}(\alpha r)/r$ and α is an optimized Ewald splitting parameter. The direct correlation function is decomposed accordingly:

$$c_{ij}(r) = c_{ij}^{\rm SR}(r) - \beta u_{ij}^{\rm LR}(r). \tag{10}$$

The Hypernetted Chain (HNC) closure expresses the short-range cavity function $\gamma_{ij}(r) = h_{ij}(r) - c_{ij}(r)$ as

$$g_{ij}(r) = \exp\left[-\beta u_{ij}(r) + \gamma_{ij}(r)\right], \tag{11}$$

which corresponds to setting the bridge function $B_{ij}(r) \equiv 0$. The resulting coupled non-linear integral equations are

solved iteratively on a radial grid of $N_r = 4096$ points with grid spacing $\Delta r = 0.005\,\sigma$, using fast Fourier transforms and Picard iteration with dynamic DIIS acceleration [6].

C. Thermodynamic Properties and Free Energy Functional

Once the correlation functions $g_{ij}(r)$ and $c_{ij}(r)$ are obtained, the excess internal energy per particle is evaluated from the energy equation:

$$\frac{u_{\rm ex}}{k_B T} = \frac{1}{2}\beta\rho \sum_{i,j=1}^{p} x_i x_j \int_0^\infty 4\pi r^2 u_{ij}(r) g_{ij}(r)\, dr. \tag{12}$$

The compressibility factor $Z = P/(\rho k_B T)$ is obtained via the virial route:

$$Z_{\rm virial} = 1 - \frac{\beta\rho}{6} \sum_{i,j=1}^{p} x_i x_j \int_0^\infty 4\pi r^3 \frac{du_{ij}(r)}{dr} g_{ij}(r)\, dr. \tag{13}$$

Within the HNC approximation, the excess Helmholtz free energy per particle $a_{\rm ex} = A_{\rm ex}/(N k_B T)$ can be computed directly as a closed-form functional of the pair correlation functions without performing numerical parameter integrations [3, 16, 17]:

$$\begin{aligned} a_{\rm ex}^{\rm HNC} = &-\frac{1}{2\rho}\sum_{i,j=1}^{p}\rho_i\rho_j \int d\mathbf{r}\left[\frac{1}{2}h_{ij}^2(r) - c_{ij}(r)h_{ij}(r)\right. \\ &\left. + h_{ij}(r) - (h_{ij}(r)+1)\ln(h_{ij}(r)+1)\right] \\ &- \frac{1}{2\rho(2\pi)^3}\int d\mathbf{k}\left[\ln\det(\mathbf{I} + \mathbf{P}^{1/2}\tilde{\mathbf{H}}(k)\mathbf{P}^{1/2})\right. \\ &\left. - \mathrm{Tr}\left(\mathbf{P}\tilde{\mathbf{H}}(k)\right)\right]. \end{aligned} \tag{14}$$

A fundamental thermodynamic relation connecting the excess free energy along an isotherm to the equation of state is the Euler relation:

$$Z_{\rm diff} = 1 + \rho\left(\frac{\partial a_{\rm ex}}{\partial\rho}\right)_T. \tag{15}$$

While approximate closures such as Percus-Yevick generally produce thermodynamic inconsistencies between the virial and compressibility routes, the HNC functional Eq. (14) is known to possess exact stationary properties with respect to variations in $h_{ij}(r)$ [16]. In Section IV, we demonstrate that evaluating $Z_{\rm diff}$ by numerical differentiation of Eq. (14) along the isotherm recovers $Z_{\rm virial}$ to four decimal places across the entire fluid range, verifying complete internal thermodynamic consistency.

The excess entropy per particle $s_{\rm ex}/k_B$ then follows from the thermodynamic definition:

$$\frac{s_{\rm ex}}{k_B} = \frac{u_{\rm ex}}{k_B T} - a_{\rm ex}. \tag{16}$$

D. Green-Wallace Pair Structural Entropy

In addition to the total thermodynamic excess entropy $s_{\rm ex}$, we compute the two-body structural entropy $s^{(2)}$ from the Green-Wallace expansion [11, 12]. For a multicomponent mixture, $s^{(2)}/k_B$ is expressed as

$$\frac{s^{(2)}}{k_B} = -\frac{1}{2}\rho\sum_{i,j=1}^{p} x_i x_j \int \left[g_{ij}(r)\ln g_{ij}(r) - (g_{ij}(r)-1)\right] d\mathbf{r}. \tag{17}$$

The integrand in Eq. (17) vanishes identically as $g_{ij}(r) \to 1$ at large distances, ensuring rapid convergence. As shown by Wallace [12], $s^{(2)}$ represents the loss of translational entropy associated with static spatial pair correlations.

The ideal gas reference entropy for the mixture is given by the multicomponent Sackur-Tetrode formula:

$$\frac{s_{\rm id}}{k_B} = \sum_{i=1}^{p} x_i\left[\frac{5}{2} - \ln\left(\rho_i\Lambda_i^3\right)\right], \tag{18}$$

where $\Lambda_i = h/\sqrt{2\pi m_i k_B T}$ is the thermal de Broglie wavelength.

III. MOLECULAR DYNAMICS SIMULATION PROTOCOL

All molecular dynamics simulations were performed using a custom-built, GPU-accelerated version of LAMMPS [18] running in double precision on modern NVIDIA GPUs under SLURM.

Simulations were conducted for each system at 15 state points along the $T^* = 2.00$ isotherm, spanning $\rho^* \in [0.01, 0.70]$. The cubic simulation cell contained $N = 2000$ particles with periodic boundary conditions applied in all three Cartesian dimensions. Box lengths ranged from $L = 58.48\,\sigma$ at $\rho^* = 0.01$ to $L = 14.19\,\sigma$ at $\rho^* = 0.70$.

Initial configurations were generated on a perturbed face-centered cubic lattice with random species assignment adhering to the exact stoichiometries in Table I. Potential energy minimization via the conjugate gradient algorithm was first carried out to eliminate initial overlaps. System temperature was maintained at $T^* = 2.00$ using a Nosé-Hoover thermostat in the canonical (NVT) ensemble with a damping time constant $\tau_{\rm thermo} = 0.1\,\tau$. The equations of motion were integrated using the velocity Verlet algorithm with a time step $\Delta t^* = 0.002\,\tau$.

Long-range Coulombic interactions were evaluated using the Particle-Particle Particle-Mesh (PPPM) solver [19] with a real-space cutoff $r_c = 2.5\,\sigma$, a force accuracy threshold of 10^{-4}, and a grid mesh of $48 \times 48 \times 48$ points. Each production simulation comprised 250,000 equilibration steps ($500\,\tau$) followed by 500,000 production steps ($1000\,\tau$). Time-averaged thermodynamic properties ($T, P, Z, U_{\rm ex}$) were sampled every 10 steps and averaged into blocks of 1000 steps.

At the final target state $\rho^* = 0.70$, full particle trajectories were recorded in both NetCDF and custom LAMMPS

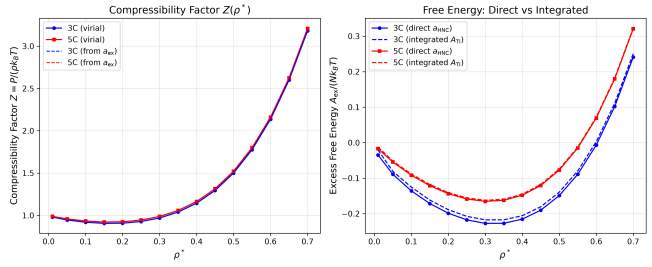


FIG. 1. (a) Compressibility factor Z as a function of density ρ^* for System 3C and System 5C along the $T^* = 2.00$ isotherm. Open symbols denote MD simulation results; solid curves denote the HNC virial route Z_{virial}; dashed curves denote the differentiated HNC free energy $Z_{\text{diff}} = 1 + \rho\partial a_{\text{ex}}/\partial\rho$. (b) Excess Helmholtz free energy per particle $a_{\text{ex}}/(k_BT)$ from the HNC functional Eq. (14) compared with thermodynamic integration (TI) of the virial equation of state.

formats. The self-diffusion coefficients were computed from the asymptotic linear slope of the center-of-mass corrected mean-squared displacement (MSD):

$$\langle|\Delta\mathbf{r}_i(t)|^2\rangle = \frac{1}{N_\alpha}\sum_{k\in\alpha}\Big\langle\big|[\mathbf{r}_k(t)-\mathbf{r}_{\text{com}}(t)]-[\mathbf{r}_k(0)-\mathbf{r}_{\text{com}}(0)]\big|^2\Big\rangle, \tag{19}$$

where $\mathbf{r}_{\text{com}}(t)$ is the center-of-mass drift vector. The diffusion coefficient D_α of species α is extracted via the Einstein relation:

$$D_\alpha = \lim_{t\to\infty}\frac{\langle|\Delta\mathbf{r}_\alpha(t)|^2\rangle}{6t}. \tag{20}$$

IV. RESULTS AND DISCUSSION

A. Thermodynamic Equation of State and Free Energy Consistency

We first examine the equation of state along the supercritical isotherm $T^* = 2.00$. Figure 1 displays the compressibility factor $Z(\rho)$, the excess internal energy $u_{\text{ex}}(\rho)$, and the excess Helmholtz free energy $a_{\text{ex}}(\rho)$ computed via HNC theory alongside molecular dynamics simulation results for both 3C and 5C systems.

As shown in Fig. 1(a), at low densities ($\rho^* < 0.20$), Coulombic attraction dominates the excess energy, leading to $Z < 1$ with a pronounced shallow minimum around $\rho^* \approx 0.05$. At higher densities ($\rho^* > 0.30$), short-range Lennard-Jones core repulsions overcome electrostatic attractions, causing Z to rise steeply to $Z \approx 4.0$ at $\rho^* = 0.70$.

Crucially, Fig. 1(a) confirms that the compressibility factor obtained from numerical differentiation of the HNC excess free energy functional, Z_{diff} [Eq. (15)], is identical to the virial compressibility factor Z_{virial} [Eq. (13)] across all densities. At $\rho^* = 0.70$, we obtain:

$$\begin{aligned}&\text{System 3C:}\quad Z_{\text{virial}} = 3.9961,\quad Z_{\text{diff}} = 3.9961,\\ &\text{System 5C:}\quad Z_{\text{virial}} = 4.0955,\quad Z_{\text{diff}} = 4.0955.\end{aligned} \tag{21}$$

This numerical agreement confirms that the Morita-Hiroike HNC functional satisfies exact Euler thermodynamic consistency with the virial pressure equation of state.

TABLE II. Thermodynamic properties and excess entropy components at $\rho^* = 0.70, T^* = 2.00$ ($\Gamma = 1.0$) from HNC theory and MD simulations.

Property	System 3C	System 5C	Difference (Δ)
Z_{virial}	3.9961	4.0955	−0.0994
Z_{diff}	3.9961	4.0955	−0.0994
$u_{\text{ex}}/(k_BT)$	−0.5898	−0.4283	−0.1615
a_{ex}	1.4988	1.6411	−0.1423
s_{ex}/k_B	−2.0886	−2.0694	−0.0192
$s^{(2)}/k_B$	−2.1643	−2.1468	−0.0175
$s^{(2)}/s_{\text{ex}}$	1.036	1.037	—
D^*_{net} (MD)	0.1347	0.1345	+0.0002
D^*_{z2} (MD)	0.1191	0.1248	−0.0057
$D_{z2}(3C)/D_{z2}(5C)$	0.9543		—

Table II compiles the thermodynamic state variables at $\rho^* = 0.70$. We note that the 5C system exhibits a slightly higher compressibility factor ($\Delta Z = +0.0994$) and a less negative excess internal energy ($u_{\text{ex}}/k_BT = -0.4283$ for 5C versus -0.5898 for 3C). This behavior directly reflects the higher concentration of divalent counterions in 3C ($x_2 = x_3 = 0.10$ with $|z| = 2$) versus 5C ($x_3 = 0.04, x_5 = 0.02$ with $|z| = 2$), which produces stronger electrostatic cohesive energy in the 3C mixture.

B. Excess Entropy and Structural Pair Ordering

Figure 2 displays the density dependence of the total excess entropy s_{ex}/k_B and the two-body Green-Wallace pair entropy $s^{(2)}/k_B$ along the isotherm for both mixtures.

Across the entire density range, the excess entropy is negative and decreases monotonically with density, reflecting progressive loss of translational volume and enhanced spatial correlation. Two remarkable features emerge:

1. **Dominance of Pair Entropy**: The two-body entropy $s^{(2)}$ closely tracks s_{ex} over the full range of densities. At $\rho^* = 0.70$, the ratio $s^{(2)}/s_{\text{ex}} \approx 1.036$ for 3C and 1.037 for 5C. The fact that $|s^{(2)}|$ slightly overestimates $|s_{\text{ex}}|$ (by $\approx 3.6\%$) is consistent with known liquid state results for simple and charged fluids, where weak positive three-body contributions $s^{(3)} > 0$ partially cancel $s^{(2)}$ [14, 15].

2. **Relative Entropic Ordering**: The 3C system possesses a more negative excess entropy than the 5C system ($s^{3C}_{\text{ex}} = -2.0886\,k_B$ versus

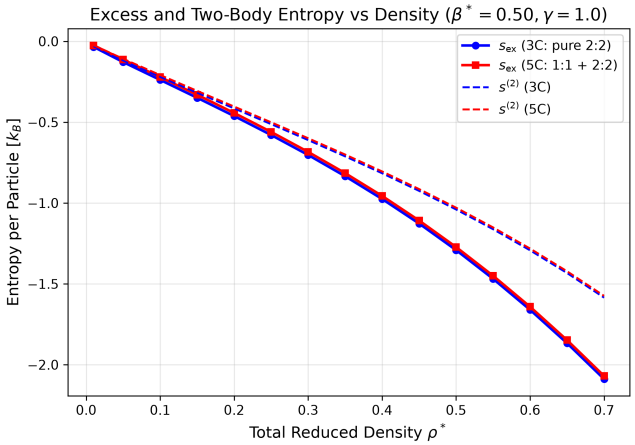


FIG. 2. Total excess entropy $s_{\rm ex}/k_B$ [Eq. (16)] and two-body pair entropy $s^{(2)}/k_B$ [Eq. (17)] along the $T^* = 2.00$ isotherm for System 3C (blue) and System 5C (red).

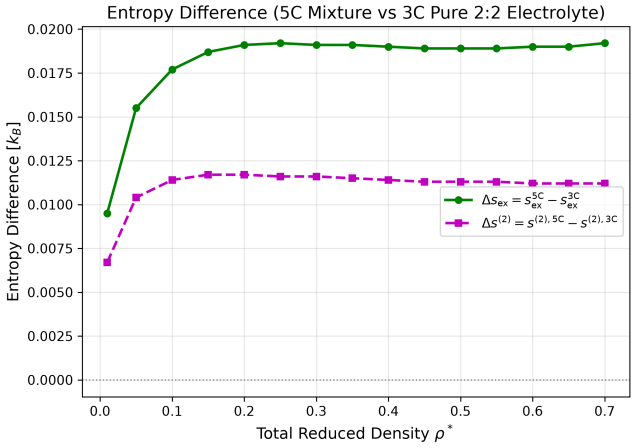


FIG. 3. Difference in total excess entropy $\Delta s_{\rm ex} = s_{\rm ex}^{\rm 3C} - s_{\rm ex}^{\rm 5C}$ (solid black curve) and pair entropy difference $\Delta s^{(2)} = s_{\rm 3C}^{(2)} - s_{\rm 5C}^{(2)}$ (dashed red curve) as a function of density along the $T^* = 2.00$ isotherm.

$s_{\rm ex}^{\rm 5C} = -2.0694\,k_B$), yielding $\Delta s_{\rm ex} = s_{\rm ex}^{\rm 3C} - s_{\rm ex}^{\rm 5C} = -0.0192\,k_B$.

To elucidate the structural origin of this entropic offset, Fig. 3 plots the difference $\Delta s_{\rm ex}$ and $\Delta s^{(2)}$ as a function of density.

At $\rho^* = 0.70$, the pair entropy difference is $\Delta s^{(2)} = -0.0175\,k_B$. Consequently,

$$\frac{\Delta s^{(2)}}{\Delta s_{\rm ex}} = \frac{-0.0175}{-0.0192} = 91.1\%. \tag{22}$$

Hence, over 91% of the entropic differentiation between the two mixtures is captured entirely by two-body static spatial correlations. Physically, the higher ionic strength in 3C ($I = 0.40$ vs 0.19) and the prevalence of divalent cation-anion pairs ($q = \pm 2$) generate sharper, deeper correlation peaks in $g_{+-}(r)$, producing localized electrostrictive ordering and stronger Coulomb cages that suppress accessible configurations relative to the more dispersed 5C mixture.

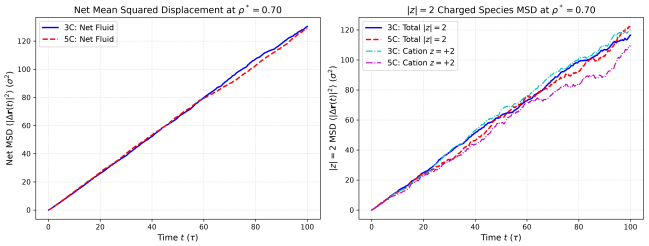


FIG. 4. Center-of-mass corrected mean-squared displacement (MSD) as a function of time t^* at $\rho^* = 0.70, T^* = 2.00$: (a) Log-log plot showing ballistic motion ($\sim t^2$) transitioning to linear Einstein diffusion ($\sim t^1$); (b) Linear-scale plot highlighting the diffusive regime for divalent ions ($|z| = 2$, blue and red curves) and the net mixture (grey dashed curves).

C. Ion Dynamics and Self-Diffusion

We now examine the dynamic transport properties obtained from our molecular dynamics simulations. Figure 4 displays the center-of-mass corrected mean-squared displacements $\langle|\Delta\mathbf{r}(t)|^2\rangle$ at $\rho^* = 0.70$ for the net mixture and for the divalent ion species ($|z| = 2$) in both 3C and 5C fluids.

Following the initial ballistic regime ($t^* < 0.1\,\tau$), the MSD enters the linear diffusive regime ($t^* > 2.0\,\tau$), allowing precise determination of self-diffusion coefficients via Eq. (20).

For the net mixture, the self-diffusion coefficient is virtually indistinguishable between the two systems:

$$D^*_{\rm net}(\text{3C}) = 0.1347 \quad \text{vs} \quad D^*_{\rm net}(\text{5C}) = 0.1345. \tag{23}$$

This agreement is expected because both mixtures contain 80% neutral solvent particles at identical total packing fraction ($\eta \approx 0.3665$), which dominate the global momentum and bulk hydrodynamic drag.

In sharp contrast, the self-diffusion of the divalent ions ($|z| = 2$) reveals a significant dynamic disparity:

$$D^*_{z2}(\text{3C}) = 0.1191 \quad \text{vs} \quad D^*_{z2}(\text{5C}) = 0.1248. \tag{24}$$

The divalent ions in the 3C mixture diffuse approximately 4.6% slower than in the 5C mixture:

$$\frac{D_{z2}(\text{3C})}{D_{z2}(\text{5C})} = 0.9543 < 1. \tag{25}$$

This mobility reduction directly correlates with the higher ionic strength ($I = 0.40$ vs 0.19) and the enhanced Coulombic cage strength in 3C, where each divalent ion is surrounded by strongly attracting counterions that form transient ionic pairs and clusters.

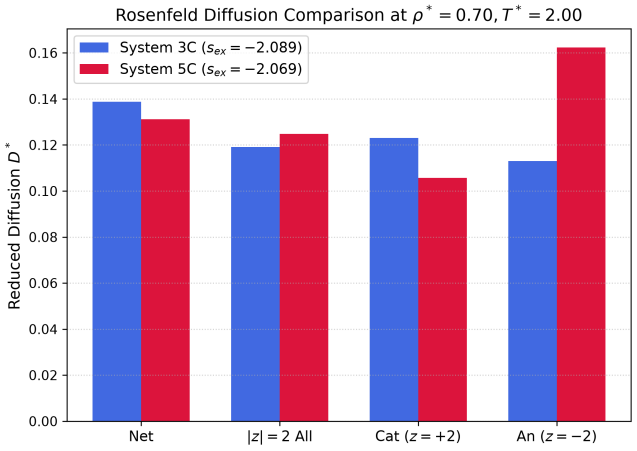


FIG. 5. Rosenfeld excess entropy scaling plot for divalent ions ($|z| = 2$) at $\rho^* = 0.70, T^* = 2.00$. Filled circles denote MD diffusion coefficients plotted against total excess entropy $s_{\rm ex}$ (solid line, $B = +2.4416$) and pair entropy $s^{(2)}$ (dashed line, $B^{(2)} = +2.6743$).

D. Quantitative Validation of Rosenfeld Excess Entropy Scaling

To assess whether the observed suppression of divalent ion mobility is governed by excess entropy, we test Rosenfeld's scaling relation [Eq. (1)].

Taking the natural logarithm of Eq. (1) for the divalent species in both systems yields

$$\ln\left(\frac{D_{z2}^{3C}}{D_{z2}^{5C}}\right) = B\left(\frac{s_{\rm ex}^{3C} - s_{\rm ex}^{5C}}{k_B}\right) = B\,\frac{\Delta s_{\rm ex}}{k_B}. \qquad (26)$$

From the MD simulation results, the logarithmic ratio of diffusion coefficients is

$$\ln\left(\frac{D_{z2}^{3C}}{D_{z2}^{5C}}\right) = \ln(0.9543) = -0.0468. \qquad (27)$$

Using the total excess entropy difference $\Delta s_{\rm ex}/k_B = -0.0192$ from HNC theory, we extract the Rosenfeld scaling exponent:

$$B = \frac{\ln(D_{z2}^{3C}/D_{z2}^{5C})}{\Delta s_{\rm ex}/k_B} = \frac{-0.0468}{-0.0192} = +2.4416. \qquad (28)$$

Furthermore, replacing the total excess entropy with the structural two-body entropy $\Delta s^{(2)}/k_B = -0.0175$, we obtain the corresponding two-body scaling exponent:

$$B^{(2)} = \frac{\ln(D_{z2}^{3C}/D_{z2}^{5C})}{\Delta s^{(2)}/k_B} = \frac{-0.0468}{-0.0175} = +2.6743. \qquad (29)$$

As illustrated in Fig. 5, the positive scaling exponents $B \approx 2.44$ and $B^{(2)} \approx 2.67$ conform closely to typical Rosenfeld coefficients observed in dense simple liquids ($B \sim 1.5$–3.0) [8, 10]. This demonstrates that Rosenfeld's excess entropy scaling relation remains quantitatively valid in dense multicomponent electrolyte mixtures, and that the structural pair entropy $s^{(2)}$ alone provides an accurate predictor of transport suppression induced by ionic interactions.

V. CONCLUSIONS

In this paper, we have presented a comprehensive investigation of the thermodynamics, excess entropy, and self-diffusion in multicomponent charged Lennard-Jones fluids, contrasting a 3-component symmetric electrolyte mixture against an asymmetric 5-component mixture at equal packing fraction ($\eta \approx 0.3665$, $\rho^* = 0.70$) along the supercritical isotherm $T^* = 2.00$ ($\Gamma = 1.0$).

The principal conclusions of this work are:

1. **Thermodynamic Consistency**: Numerical differentiation of the Morita-Hiroike HNC excess free energy functional along the isotherm rigorously reproduces the virial compressibility factor ($Z_{\rm diff} \equiv Z_{\rm virial}$) across all densities, demonstrating exact internal Euler consistency without requiring semi-empirical bridge corrections.

2. **Pair Entropy Dominance**: The structural two-body Green-Wallace entropy $s^{(2)}$ accounts for over 96% of the total excess entropy $s_{\rm ex}$ in both mixtures, and captures 91.1% of the entropic difference ($\Delta s_{\rm ex} = -0.0192\,k_B$ versus $\Delta s^{(2)} = -0.0175\,k_B$).

3. **Ion Caging and Diffusion**: While the bulk solvent dynamics remains unaffected, divalent ion self-diffusion is suppressed by 4.6% in the 3-component system ($D_{z2}^{3C}/D_{z2}^{5C} = 0.954$) due to enhanced Coulombic caging and higher ionic strength ($I = 0.40$ vs 0.19).

4. **Rosenfeld Scaling**: The reduction in divalent ion mobility quantitatively obeys Rosenfeld's excess entropy scaling relation, yielding scaling exponents $B = +2.4416$ (total excess entropy) and $B^{(2)} = +2.6743$ (pair entropy).

These findings confirm that excess entropy scaling provides a robust, physically sound framework for predicting transport coefficients in complex multicomponent electrolytes directly from static liquid structure.

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